Data Mining and Data Analytics - R Lab Record

Digital, copy-friendly version of your handwritten record (pages 53-106) for exam revision.

All code is in monospace boxes so it can be selected and pasted straight into RStudio. Handwriting slips in the code (e.g. 'Gov' for aov, 'list(' for hist(, 'mean=' for main=, missing brackets) have been corrected so that it actually runs. Outputs are copied from your record; where I spotted a number that doesn't match what R would produce, it is flagged in the last section.

1	Mean, Median, Mode	7	Time Series Analysis
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			Corrections / things to verify (last page)

1. Mean, Median, Mode

Record p. 53

R code (copy-paste ready)

```
x <- c(10, 20, 30, 20, 40, 50, 20)
mean_value   <- mean(x)
median_value <- median(x)
unique_values <- unique(x)
mode_value <- unique_values[which.max(tabulate(match(x, unique_values)))]

cat("Mean:",   mean_value,   "\n")
cat("Median:", median_value, "\n")
cat("Mode:",   mode_value,   "\n")
```

Output (from your record)

```
Mean: 27.14286
Median: 20
Mode: 20
```

Note: Your record says 27.17286; the correct mean of these numbers is 190/7 = 27.14286.

Mode trick: match() gives each value's position in unique_values, tabulate() counts how often each position occurs, which.max() picks the most frequent one.

2. ANCOVA - Regression Visualization (mtcars)

Record pp. 53-55

R code (copy-paste ready)

```
input <- mtcars

# Model 1: with interaction (hp * am)
result1 <- aov(mpg ~ hp * am, data = input)
print(summary(result1))

# Model 2: without interaction (hp + am)
result2 <- aov(mpg ~ hp + am, data = input)
print(summary(result2))

# Compare the two models
print(anova(result1, result2))

library(ggplot2)
ggplot(input, aes(x = hp, y = mpg, color = factor(am))) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "ANCOVA Regression Lines",
```

```
       x = "Horse power",
       y = "Miles per Gallon",
       color = "Transmission")
```

Output (from your record)

```
# summary(result1)   mpg ~ hp * am
            Df Sum Sq Mean Sq F value   Pr(>F)
hp           1  678.4   678.4  77.391 1.50e-09 ***
am           1  202.2   202.2  23.072 4.78e-05 ***
hp:am        1    0.0     0.0   0.001    0.981
Residuals   28  245.4     8.8
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# summary(result2)   mpg ~ hp + am
            Df Sum Sq Mean Sq F value   Pr(>F)
hp           1  678.4   678.4   80.15 7.63e-10 ***
am           1  202.2   202.2   23.89 3.46e-05 ***
Residuals   29  245.4     8.5
---

# anova(result1, result2)
  Res.Df    RSS Df  Sum of Sq      F Pr(>F)
1     28 245.43
2     29 245.44 -1 -0.0052815  6e-04 0.9806
```

*Note: The anova() call is not written in your code but its output is in your record, so I added it. Model 1 is hp*am and Model 2 is hp+am (your record writes hp+am for both).*

Plot: scatter of hp (x) vs mpg (y), points coloured by transmission (am = 0/1), with two roughly parallel downward-sloping regression lines.

Interpretation: hp and am are both significant. The hp:am interaction (p = 0.981) is not significant, so the two lines are parallel and the simpler additive model is enough.

3. Linear Regression

Record pp. 60-61

R code (copy-paste ready)

```
x <- c(141, 134, 178, 156, 108, 116, 119, 143, 162, 130)
y <- c(62, 85, 56, 21, 47, 17, 76, 92, 62, 58)

relationship_model <- lm(y ~ x)
print(summary(relationship_model))

plot(x, y,
     col  = "red",
     main = "height and weight Regression",
     cex  = 1.3,
     pch  = 16,
     xlab = "weight in kg",
     ylab = "Height in cm")
abline(relationship_model)
```

Output (from your record)

```
Call:
lm(formula = y ~ x)

Residuals:
    Min      1Q  Median      3Q     Max
-38.948  -7.390   1.869  15.933  34.087

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 47.50833   55.98118   0.849    0.421
x            0.07276    0.39342   0.185    0.858

Residual standard error: 25.96 on 8 degrees of freedom
Multiple R-squared:  0.004257,  Adjusted R-squared:  -0.1202
F-statistic: 0.0342 on 1 and 8 DF,  p-value: 0.8579
```

Note: Your record has intercept t = 0.867 / p = 0.414; 47.50833/55.98118 = 0.849 (p about 0.42). Everything else matches.

Interpretation: R-squared is about 0.4% and p = 0.858, so x is not a useful predictor of y here (almost no linear relationship).

4. Multiple Regression (mtcars)

Record pp. 62-63

R code (copy-paste ready)

```
input <- mtcars[, c("mpg", "wt", "disp", "hp")]

model <- lm(mpg ~ wt + disp + hp, data = input)
print(model)
print(summary(model))

predicted <- predict(model)

plot(input$mpg, predicted,
     main = "Multiple Regression: Actual vs predicted MPG",
     xlab = "Actual MPG",
     ylab = "Predicted MPG",
     col  = "blue",
     pch  = 16)
abline(0, 1, col = "red", lwd = 2)
```

Output (from your record)

```
Call:
lm(formula = mpg ~ wt + disp + hp, data = input)

Coefficients:
(Intercept)           wt         disp           hp
  37.105505    -3.800891    -0.000937    -0.031157

Residuals:
    Min      1Q  Median      3Q     Max
-3.891  -1.640  -0.182   1.061   5.861

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 37.105505   2.110815  17.579  < 2e-16 ***
wt          -3.800891   1.066191  -3.565  0.00133 **
disp        -0.000937   0.010350  -0.091  0.92851
hp          -0.031157   0.011436  -2.724  0.01097 *
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.639 on 28 degrees of freedom
Multiple R-squared:  0.8268,  Adjusted R-squared:  0.8083
F-statistic: 44.57 on 3 and 28 DF,  p-value: 8.65e-11
```

Note: Small slips in your record (wt coefficient written -3.800981, wt SE 1.66191, disp 0.00937, median residual unclear) were corrected to the values R gives.

Plot: actual MPG (x) vs predicted MPG (y); points hug the red y = x line, showing a good fit.

Interpretation: R-squared 0.83 means the model explains about 83% of the variation in mpg. wt (p = 0.001) and hp (p = 0.011) are significant; disp is not (p = 0.93).

5. Logistic Regression + ROC Curve

Record pp. 63-65

R code (copy-paste ready)

```
claimants <- read.csv(file.choose())
claimants <- na.omit(claimants)

logit <- glm(ATTORNEY ~ factor(CLMSEX) + factor(CLMINSUR) + factor(SEATBELT) +
               CLMAGE + LOSS,
             family = "binomial",
             data   = claimants)
```

```
print(summary(logit))

prob <- predict(logit, newdata = claimants, type = "response")
print(prob)

# Confusion matrix and accuracy (threshold 0.5)
confusion <- table(prob > 0.5, claimants$ATTORNEY)
print(confusion)
Accuracy <- sum(diag(confusion)) / sum(confusion)
print(Accuracy)

# ROC curve
library(ROCR)
library(pROC)
rocpred <- prediction(prob, claimants$ATTORNEY)
rocperf <- performance(rocpred, "tpr", "fpr")
plot(rocperf,
     colorize = TRUE,
     main = "ROC curve - logistic regression")

auc_value <- auc(claimants$ATTORNEY, prob)
print(auc_value)
```

Output (from your record)

```
Call:
glm(formula = ATTORNEY ~ factor(CLMSEX) + factor(CLMINSUR) + factor(SEATBELT) +
    CLMAGE + LOSS, family = "binomial", data = claimants)

Coefficients:
                   Estimate Std. Error z value Pr(>|z|)
(Intercept)      -1.566e+01  4.667e+03       0        1
factor(CLMSEX)2   1.576e+00  3.123e+05       0        1
factor(CLMINSUR)2 -2.616e+01 2.169e+05       0        1
factor(SEATBELT)2 -2.689e+01 1.043e+06       0        1
CLMAGE             5.599e+00  7.303e+04      0        1
LOSS              -3.291e+03  1.455e+02       0        1   (digits hard to read in scan)

(Dispersion parameter for binomial family taken to be 1)
    Null deviance: 2.7726e+01  on 19  degrees of freedom
Residual deviance: 3.8111e-10  on 14  degrees of freedom
AIC: 12

Number of Fisher Scoring iterations: 25

# table(prob > 0.5, claimants$ATTORNEY)
         0  1
  FALSE 10  0
  TRUE   0 10

Setting levels: control = 0, case = 1
Setting direction: controls < cases
Area under the curve: 1
```

Plot: ROC curve (False positive rate vs True positive rate) jumps straight up to 1 - a perfect classifier, AUC = 1.

Interpretation: with only 20 rows the model separates the classes perfectly (huge standard errors, residual deviance about 0, 25 Fisher iterations). This is complete separation / overfitting, so the AUC of 1 should not be trusted as real-world performance.

Note: Your record writes the packages as 'ROCR' and 'PROCR' - the real names are ROCR and pROC. Install with install.packages(c('ROCR','pROC')) if needed.

6. Poisson Regression (warpbreaks)

Record pp. 66-69

R code (copy-paste ready)

```
input <- warpbreaks
print(head(input))

output <- glm(breaks ~ wool + tension,
              data   = input,
```

```
              family = poisson)
print(output)
print(summary(output))

predicted <- predict(output, type = "response")
print(predicted)

plot(input$breaks, predicted,
     main = "Poisson Regression: Actual vs predicted breaks",
     xlab = "Actual no of breaks",
     ylab = "Predicted no of breaks",
     col  = "blue",
     pch  = 16)
abline(0, 1, col = "red", lwd = 2)
```

Output (from your record)

```
  breaks wool tension
1     26    A       L
2     30    A       L
3     54    A       L
4     25    A       L
5     70    A       L
6     52    A       L

Call:  glm(formula = breaks ~ wool + tension, family = poisson, data = input)

Coefficients:
(Intercept)        woolB     tensionM     tensionH
     3.6920      -0.2060      -0.3213      -0.5185

Degrees of Freedom: 53 Total (i.e. Null);  50 Residual
Null Deviance:     297.4
Residual Deviance: 210.4    AIC: 493.1

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   3.69196    0.04541  81.302  < 2e-16 ***
woolB        -0.20599    0.05157  -3.994 6.49e-05 ***
tensionM     -0.32132    0.06027  -5.332 9.73e-08 ***
tensionH     -0.51849    0.06396  -8.107 5.21e-16 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note: Your record shows intercept 3.66196 and woolB -0.660 in places; the correct values are 3.69196 and -0.2060 (R output).

Plot: actual breaks (x, 10 to 70) vs predicted breaks (y, roughly 20 to 40). Predictions are compressed - the model under-predicts large counts.

Interpretation: all predictors are significant. Wool B and higher tension (M, H) both reduce the expected number of breaks. Poisson uses a log link, so exp(coef) gives the multiplicative effect, e.g. exp(-0.206) is about 0.81 (19% fewer breaks for wool B).

7. Time Series Analysis

Record pp. 74-76

R code (copy-paste ready)

```
snowfall <- c(790, 1170.8, 860.1, 1330.6, 630.4, 911.5,
              683.5, 996.6, 783.2, 982.0, 881.8, 1021)

snowfall_ts <- ts(snowfall, start = c(2013, 1), frequency = 12)
print(snowfall_ts)

plot(snowfall_ts,
     main = "Monthly snowfall time series",
     xlab = "year",
     ylab = "snowfall",
     col  = "blue",
     lwd  = 2)

time  <- 1:length(snowfall)
model <- lm(snowfall ~ time)
```

```
abline(model, col = "red", lwd = 2)
summary(model)
```

Output (from your record)

```
       Jan    Feb   Mar    Apr   May   Jun   Jul   Aug   Sep   Oct   Nov    Dec
2013 790.0 1170.8 860.1 1330.6 630.4 911.5 683.5 996.6 783.2 982.0 881.8 1021.0

Call:
lm(formula = snowfall ~ time)

Residuals:
     Min       1Q   Median       3Q      Max
-295.35  -133.35   -15.97    91.96   401.09

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  944.514    127.613   7.401 2.31e-05 ***
time          -3.752     17.339  -0.216    0.833
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Multiple R-squared:  0.004661,  Adjusted R-squared:  -0.09487
F-statistic: 0.04683 on 1 and 10 DF,  p-value: 0.833
```

Note: Verified by recalculation: intercept 944.514 (record: 944.574), min residual -295.35 (record: -298.35), adjusted R-squared -0.09487.

Plot: monthly snowfall line with a very flat red trend line - zig-zag series, no real trend.

Interpretation: slope is -3.75 per month with p = 0.833, so there is no significant linear trend over the 12 months.

8. Non-linear Least Squares Regression (nls)

Record pp. 76-77

R code (copy-paste ready)

```
x_values <- c(1.6, 2.1, 2, 2.23, 3.71, 3.25, 3.4, 3.86, 1.19, 2.21)
y_values <- c(5.19, 7.43, 6.94, 8.11, 18.75, 14.88, 16.06, 17.12, 3.21, 7.58)

model <- nls(y_values ~ b1 * x_values^2 + b2,
             start = list(b1 = 1, b2 = 3))
summary(model)

new.data <- data.frame(x_values = seq(min(x_values), max(x_values), length.out = 100))

plot(x_values, y_values,
     main = "Non linear least squares regression",
     xlab = "x_values",
     ylab = "y_values",
     pch  = 19)
lines(new.data$x_values, predict(model, newdata = new.data))

print(sum(resid(model)^2))
print(confint(model))
```

Output (from your record)

```
Formula: y_values ~ b1 * x_values^2 + b2

Parameters:
   Estimate Std. Error t value Pr(>|t|)
b1  1.19542    0.02503  47.744 4.68e-11 ***
b2  1.9902     0.21663   9.218 1.58e-05 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3678 on 8 degrees of freedom

Number of iterations to convergence: 1
Achieved convergence tolerance: 1.082e-08

[1] 1.081935

Waiting for profiling to be done...
```

```
        2.5%    97.5%
b1 1.137701 1.253135
b2 1.4973646 2.496484
```

Note: The y_values / x_values digits in the scan are hard to read - if your own run gives different estimates, re-check those two vectors against the textbook.

Plot: scatter points lying on a smooth upward-curving (quadratic) line from about (1.2, 3) to (3.9, 17).

9. Decision Tree (C5.0 on iris)

Record pp. 77-79

R code (copy-paste ready)

```
data(iris)
install.packages("C50")      # only needed once
library(C50)

set.seed(7)
train_index <- sample(1:nrow(iris), 0.7 * nrow(iris))
training <- iris[train_index, ]
testing  <- iris[-train_index, ]

model <- C5.0(Species ~ ., data = training)
print(model)

pred <- predict(model, testing)
table(Actual = testing$Species, Predicted = pred)

accuracy <- mean(pred == testing$Species)
print(accuracy)

plot(model)
```

Output (from your record)

```
Call:
C5.0.formula(formula = Species ~ ., data = training)

Classification Tree
Number of samples: 105
Number of predictors: 4

Tree size: 4

Non-standard options: attempt to group attributes

            Predicted
Actual       setosa versicolor virginica
  setosa         14          0         0
  versicolor      0         17         1
  virginica       0          2        11

[1] 0.9333333
```

Tree structure drawn in your record (plot(model))

```
[1] Petal.Length
 |-- <= 1.9  -> Node 2  (setosa)
 |-- >  1.9  -> [3] Petal.Width
                 |-- <= 1.7 -> [4] Petal.Length
                 |            |-- <= 5 -> Node 5  (mostly versicolor)
                 |            |-- >  5 -> Node 6  (mixed versicolor / virginica)
                 |-- >  1.7 -> Node 7  (virginica)
```

Accuracy = (14 + 17 + 11) / 45 = 0.9333. Only 3 of the 45 test flowers are misclassified (1 versicolor predicted as virginica, 2 virginica predicted as versicolor).

10a. Normal Distribution: dnorm(), pnorm(), qnorm(), rnorm()

Record pp. 89-92

dnorm() - probability density

R code (copy-paste ready)

```
x <- seq(-10, 10, by = 0.1)
y <- dnorm(x, mean = 2.5, sd = 0.5)

plot(x, y, type = "l",
     main = "Normal Distribution using dnorm()",
     xlab = "x", ylab = "Density")

png(file = "dnorm.png")
plot(x, y, type = "l",
     main = "Normal Distribution using dnorm()",
     xlab = "x", ylab = "Density")
dev.off()
```

Plot: bell curve centred at 2.5, very narrow (sd = 0.5), flat near 0 elsewhere.

pnorm() - cumulative probability

R code (copy-paste ready)

```
x <- seq(-10, 10, by = 0.2)
y <- pnorm(x, mean = 2.5, sd = 2)

plot(x, y, type = "l",
     main = "Normal cumulative Distribution using pnorm()",
     xlab = "x", ylab = "Cumulative probability")

png(file = "pnorm.png")
plot(x, y, type = "l",
     main = "Normal cumulative Distribution using pnorm()",
     xlab = "x", ylab = "Cumulative probability")
dev.off()
```

Plot: S-shaped curve rising from 0 to 1, passing 0.5 at x = 2.5.

qnorm() - quantile function

R code (copy-paste ready)

```
x <- seq(0.01, 0.99, by = 0.02)
y <- qnorm(x, mean = 2, sd = 3)

plot(x, y, type = "l",
     main = "Normal Quantile Function using qnorm()",
     xlab = "probability", ylab = "Quantile")

png(file = "qnorm.png")
plot(x, y, type = "l",
     main = "Normal Quantile Function using qnorm()",
     xlab = "probability", ylab = "Quantile")
dev.off()
```

Plot: increasing curve, steep near probability 0 and 1, centred at quantile 2 for p = 0.5.

rnorm() - random numbers

R code (copy-paste ready)

```
set.seed(123)
y <- rnorm(50, mean = 2.5, sd = 0.5)

hist(y,
     main = "Random Normal distribution using rnorm()",
     xlab = "values", ylab = "frequency")

png(file = "rnorm.png")
```

```
hist(y,
     main = "Random Normal distribution using rnorm()",
     xlab = "values", ylab = "frequency")
dev.off()
```

Plot: histogram of 50 values from about 1.5 to 4, peaking around 2 to 2.5.

Note: In the record these were written as main/mean= and list(; the correct arguments are main = and hist(. Function summary: d = density, p = cumulative probability, q = quantile (inverse of p), r = random numbers.

10b. Binomial Distribution: dbinom(), pbinom(), qbinom(), rbinom()

Record pp. 93-96

dbinom() - probability mass

R code (copy-paste ready)

```
x <- seq(0, 50, by = 1)
y <- dbinom(x, size = 50, prob = 0.5)

plot(x, y, type = "h",
     main = "Binomial Distribution using dbinom()",
     xlab = "Number of success", ylab = "probability")

png(file = "dbinom.png")
plot(x, y, type = "h",
     main = "Binomial Distribution using dbinom()",
     xlab = "Number of success", ylab = "probability")
dev.off()
```

Plot: vertical-line plot, bell-shaped, peak at 25 successes.

pbinom() - cumulative probability

R code (copy-paste ready)

```
x <- 0:51
y <- pbinom(x, size = 51, prob = 0.5)

plot(x, y, type = "s",
     main = "Binomial Cumulative Distribution - pbinom()",
     xlab = "Number of heads",
     ylab = "cumulative probability")
points(26, pbinom(26, size = 51, prob = 0.5), pch = 19)
print(pbinom(26, size = 51, prob = 0.5))
```

Output (from your record)

```
[1] 0.610116
```

Plot: staircase curve rising from 0 to 1, with a solid dot at x = 26.

qbinom() - quantile function

R code (copy-paste ready)

```
p <- seq(0.01, 0.99, by = 0.01)
q <- qbinom(p, size = 51, prob = 0.5)

plot(p, q, type = "s",
     main = "Binomial Quantile function - qbinom()",
     xlab = "probability",
     ylab = "Number of heads")
points(0.25, qbinom(0.25, size = 51, prob = 0.5), pch = 19)
print(qbinom(0.25, size = 51, prob = 0.5))
```

Output (recomputed - not clearly legible in your record)

```
[1] 23
```

rbinom() - random values

R code (copy-paste ready)

```
set.seed(123)
x <- rbinom(8, size = 150, prob = 0.4)
print(x)

hist(x,
     main = "Binomial random values - rbinom()",
     xlab = "Number of success",
     ylab = "frequency")
```

Output (from your record)

```
[1] 58 60 73 53 65 57 48 65
```

Note: Your record writes 'size' as 'site' and 'sbinom' / 'obinom' in places - the function names are dbinom, pbinom, qbinom, rbinom and the argument is size.

11. Chi-square Test (Cars93: AirBags vs Type)

Record pp. 101-102

R code (copy-paste ready)

```
library(MASS)
data(Cars93)

car_data <- table(Cars93$AirBags, Cars93$Type)
print(car_data)

chi_result <- chisq.test(car_data)
print(chi_result)

barplot(car_data,
        beside = TRUE,
        col    = c("skyblue", "orange", "lightgreen"),
        legend = TRUE,
        main   = "Chi-square Test: Airbags vs Car Type",
        xlab   = "Car_Type",
        ylab   = "Frequency")
```

Output (from your record)

```
                     compact large midsize small sporty van
  Driver & Passenger       2     4       7     0      3   0
  Driver only              9     7      11     5      8   3
  None                     5     0       4    16      3   6

        Pearson's Chi-squared test

data:  car_data
X-squared = 33.001, df = 10, p-value = 0.0002723
```

Interpretation: p-value 0.00027 < 0.05, so reject H0 - airbag type and car type are NOT independent (they are associated). Note R will warn 'Chi-squared approximation may be incorrect' because some expected counts are below 5.

12a. One-sample t-test

Record pp. 103-104

R code (copy-paste ready)

```
x <- c(0.893, 0.142, 0.329, 0.691, 0.231, 0.793, 0.519, 0.392, 0.413)

t_result <- t.test(x, alternative = "greater", mu = 0.3)
print(t_result)

sample_mean <- mean(x)

hist(x,
     breaks = 5,
     col  = "lightblue",
     main = "One-sample Test",
     xlab = "Sample values",
```

```
     ylab = "Frequency")
abline(v = 0.3, col = "red", lwd = 2, lty = 2)
abline(v = sample_mean, col = "blue", lwd = 2)
legend("topright",
       legend = c("Hypothesised Mean", "Sample mean"),
       col = c("red", "blue"),
       lty = c(2, 1),
       lwd = 2)
```

Output (from your record)

```
        One Sample t-test

data:  x
t = 2.2051, df = 8, p-value = 0.02922
alternative hypothesis: true mean is greater than 0.3
95 percent confidence interval:
 0.3245133       Inf
sample estimates:
mean of x
0.4564444
```

Note: With the nine numbers as I read them, R gives mean 0.4892 and t = 2.216, p = 0.0288. Your recorded output (mean 0.4564, t = 2.2051) means one of the data values was copied differently in your notebook - please check it.

Interpretation: p = 0.029 < 0.05, so reject H0: the true mean is greater than 0.3.

12b. F-test (compare two variances)

Record pp. 105-106

R code (copy-paste ready)

```
group1 <- c(12, 15, 14, 10, 13, 16, 11, 14, 15, 13)
group2 <- c(20, 25, 22, 28, 24, 26, 21, 23, 27, 25)

f_result <- var.test(group1, group2)
print(f_result)
print(var(group1))
print(var(group2))

boxplot(group1, group2,
        names = c("Group1", "Group2"),
        col   = c("lightblue", "lightgreen"),
        main  = "F-test: comparison of variances",
        ylab  = "values")
```

Output (from your record)

```
        F test to compare two variances

data:  group1 and group2
F = 0.52709, num df = 9, denom df = 9, p-value = 0.3584
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1309226 2.1220752
sample estimates:
ratio of variances
         0.5270936

[1] 3.566667
[1] 6.766667
```

Note: I read the 9th value of group2 as 29, but your recorded variance (6.766667) only matches if it is 27, so the code above uses 27. Please check your notebook.

Interpretation: p = 0.358 > 0.05, so fail to reject H0 - no significant difference between the two variances.

Corrections and things to verify

Code fixes (handwriting vs. working R):

Where	In your record	Correct
ANCOVA	`Gov / tlabs(`	`aov / labs(`
ANCOVA	`model 1 written as hp + am`	`aov(mpg ~ hp * am)`
Logistic	`gin( / PROCR / ROCR typo`	`glm( ; library(ROCR), library(pROC)`
Normal	`mean = 'title' ; list(`	`main = 'title' ; hist(`
Binomial	`size written as site`	`size =`
Several	`missing closing brackets, mean=/main= mixups`	`fixed in this PDF`

Numbers that did not match what R would give (please compare with your notebook / re-run in RStudio):

Section	Record says	R / recalculation gives
Mean of x	27.17286	27.14286
Linear reg. intercept t, p	0.867, 0.414	0.849, about 0.42
Multiple reg. wt coef / SE	-3.800981 / 1.66191	-3.800891 / 1.066191
Poisson intercept, woolB	3.66196, -0.660	3.69196, -0.2060
Snowfall intercept, min resid	944.574, -298.35	944.514, -295.35
nls estimates	b1 1.19542, b2 1.9902	check x/y vectors (see section 8)
t-test mean / t	0.4564 / 2.2051	0.4892 / 2.216 for the data as read
F-test group2 variance	6.7667	needs 27 as 9th value (read as 29)

Quick function cheat-sheet: d* = density / probability mass, p* = cumulative probability P(X <= x), q* = quantile (inverse of p*), r* = random generation. For the normal use dnorm/pnorm/qnorm/rnorm; for the binomial use dbinom/pbinom/qbinom/rbinom.